

RIPHAH INTERNATIONAL UNIVERSITY



Faculty of Computing

FINAL YEAR PROJECT INITIAL PROPOSAL

CareLink

Project Team

Full Name of Student	SAP ID	Program	Contact Number	Email Address
Siyab Ahmed Khan	55051	BSCS	03335636526	itx.sak@gmail.com
Saif Majid Khan	57114	BSCS	03198556295	saifmajidkhan25@gmail.com
Saleem Aman	53526	BSCS	03105239909	saleemaman552@gmail.com

Mr. Muhammad Usman Sharif
(Professor)

Project Proposal

Project Title: CareLink: Agentic AI-Powered Medication Adherence & Chronic-Care Coordination System

Description:

Problem Statement

Medication non-adherence is a common problem, especially for elderly patients and people managing long-term treatments. Patients may forget doses, take medicine at the wrong time, or stop following their treatment schedule. Caregivers and family members may not know about these problems until they become serious. Nurses and care coordinators also have limited time to continuously monitor every patient. Existing medication reminder applications mainly focus on notifying patients when it is time to take their medicine. However, a reminder alone does not understand the patient's adherence behavior, determine why a patient may be missing medication, or decide when human support is needed. This creates a need for a system that can monitor medication-related information, identify changing adherence patterns, provide suitable interventions, and involve a caregiver or healthcare worker when necessary.

Proposed Solution

CareLink is suggested as a web-based healthcare support platform based on an agentic AI approach for medication adherence and care coordination. The platform will manage patient medication schedules and adherence information while using AI agents to observe patient behavior, identify adherence risks, and select appropriate interventions. Instead of treating every missed dose in the same way, the system will consider the patient's previous behavior and current situation before deciding whether to provide another reminder, notify a caregiver, or escalate the case to a nurse or care coordinator. A controlled AI assistant will also allow users to ask questions about medication and care information using trusted system data. Important AI decisions will remain explainable and subject to safety rules and human oversight. The system will follow a human-in-the-loop approach, where AI supports patients, caregivers, and care coordinators rather than replacing healthcare professionals.

Key Features

Core features include patient and medication management with medication names, doses, schedules, and adherence records. Core features also include AI-assisted analysis of medication-taking patterns to identify patients who may be at risk of poor adherence. The system will provide adaptive interventions, where reminder and follow-up strategies can change according to the patient's previous responses and adherence behavior. An agentic care-coordination workflow will prioritize important cases and notify linked caregivers or care coordinators when human attention is required. The system will provide explainable risk and intervention decisions so that users can understand why an alert or escalation was generated. A prescription/document processing component may assist in extracting medication information from uploaded prescriptions, with human verification before the information becomes part of an active medication schedule. Where a suitable wearable API is available, selected health signals such as heart rate may be used as an additional input for threshold-based caregiver alerts.

Technical Approach

The system will use a modern web-based architecture with a frontend for patients, caregivers and care coordinators, and a backend responsible for secure data and workflow management. The backend will provide APIs for patient records, medications, adherence events, alerts and care coordination. A relational database such as PostgreSQL will store critical patients and medication data. The AI layer will use multiple specialized agents, including an Adherence Agent, Intervention Agent, and Care Coordinator Agent. These agents will work with controlled tools and predefined safety rules rather than directly changing critical records. Machine learning and AI techniques will be used for adherence to pattern analysis and risk prediction. The system will also maintain explanations and logs for important AI decisions. A notification layer will handle reminders and caregiver/care-coordinator alerts. If a wearable integration is feasible, its available API will be connected through a controlled backend service. The system will follow a human-in-the-loop design, meaning that AI recommendations can be reviewed or overridden by authorized users. Critical healthcare decisions will not be left entirely to the AI system.

Objectives & Expected Outcomes

The project aims to provide a centralized system for managing medication schedules and monitoring medication adherence. The project also aims to identify changing adherence behavior earlier than a simple reminder system and provide appropriate interventions based on the patient's situation. The system aims to reduce unnecessary caregiver or nurse alerts by escalating cases according to risk and predefined safety conditions. The project also aims to demonstrate how agentic AI can coordinate multiple steps of a healthcare support workflow while maintaining explainability, safety controls, and human oversight. The platform will be delivered as an MVP over the two-semester development cycle. The implementation will focus primarily on the core

agentic adherence and care-coordination workflow, while advanced features such as extensive wearable integrations, advanced voice support and other healthcare integrations will remain optional or future work.